

MACRO L3 - 2014-2015

Lezione 6

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Session 7

These notes are those I wrote during class. They are posted "for the record".

Session 7

Nov 21, 2014

• Resource (Forest) $S(t)$

• $G(S(t))$ - natural growth rate of the forest.

• $H(t) =$ harvest rate

- rate at which humans cut trees.

(time is continuous)

$$\dot{S}(t) = \frac{dS(t)}{dt} = G(S(t)) - h(t)$$

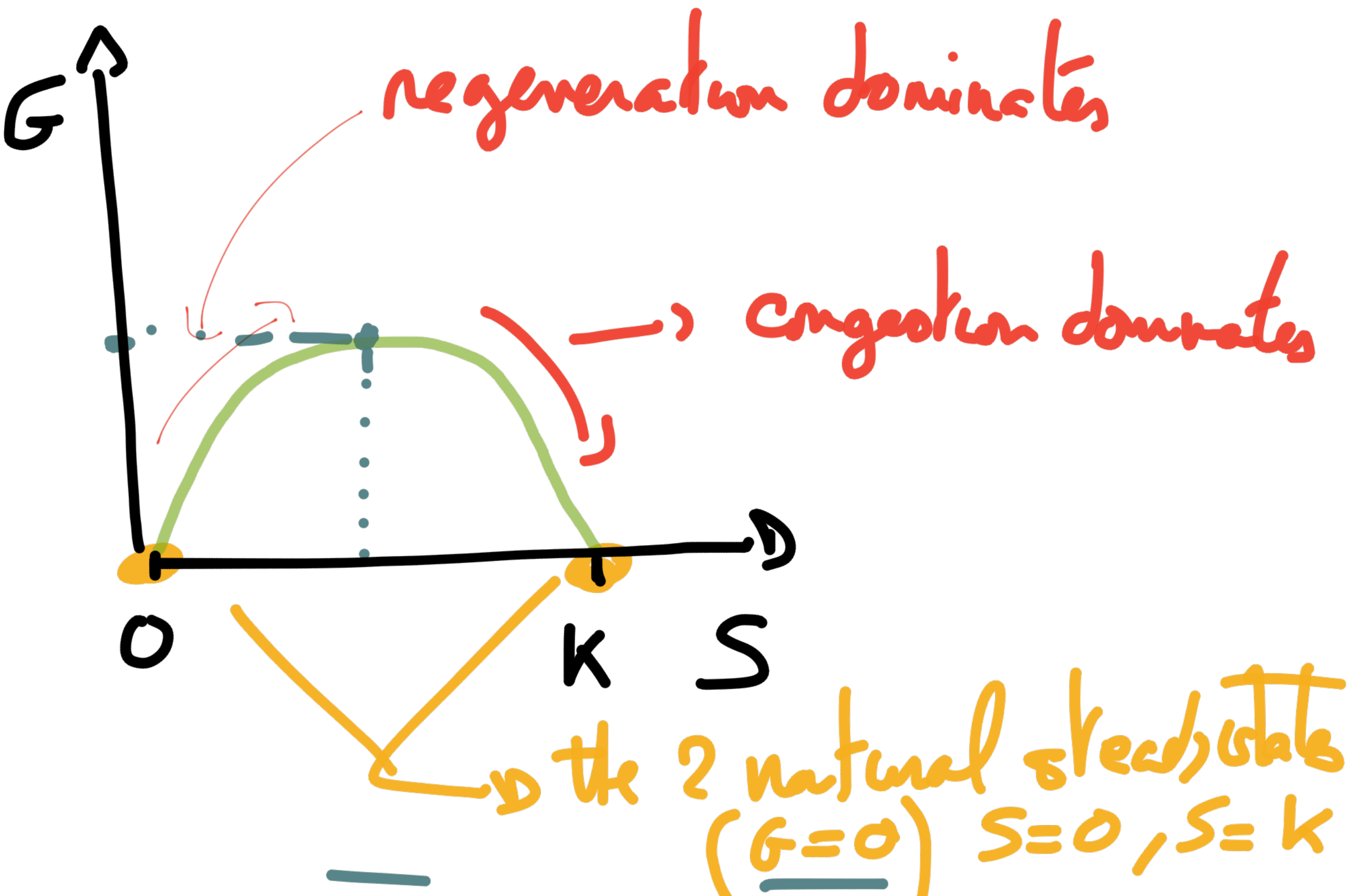
↓
= " $S(t) - S(t-dt)$ "

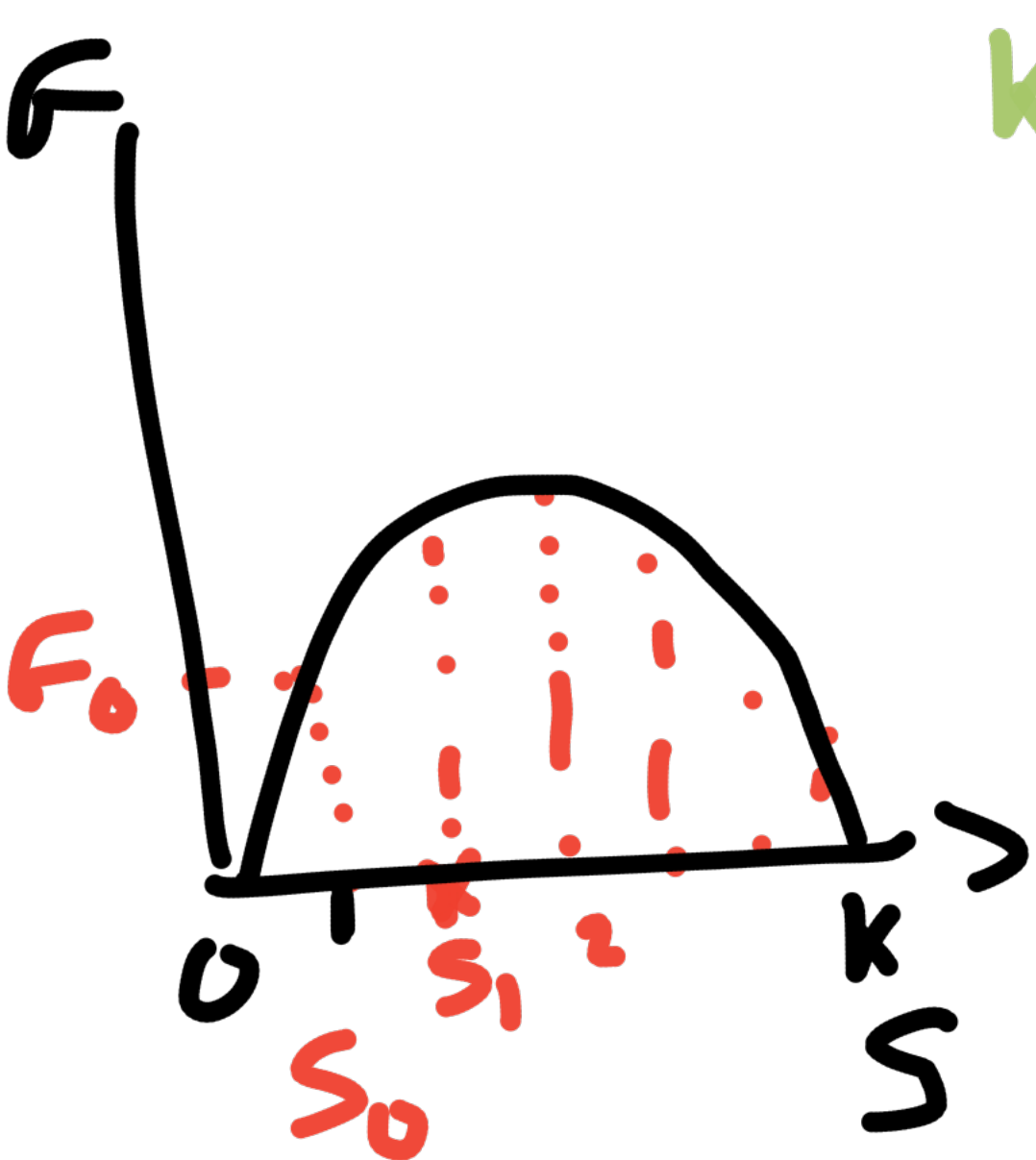
$G(S(t))$ will be be a
logistic function.

$$G(S(t)) = \underbrace{\mu S(t)} \left(1 - \underbrace{\frac{S(t)}{K}}_{\text{congestion}} \right)$$

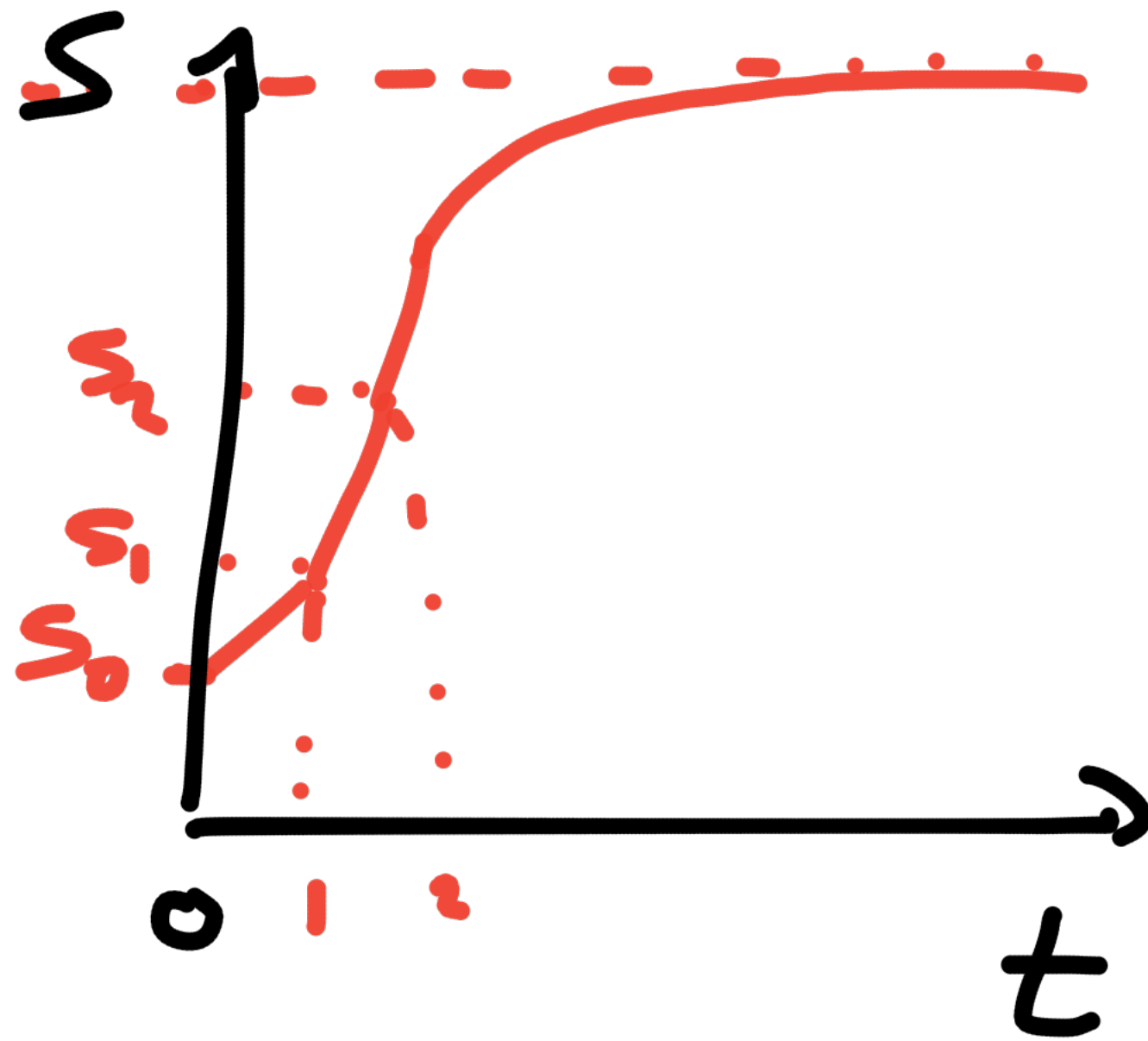
K = "carrying capacity"
 = max size of S

μ = renewal rate



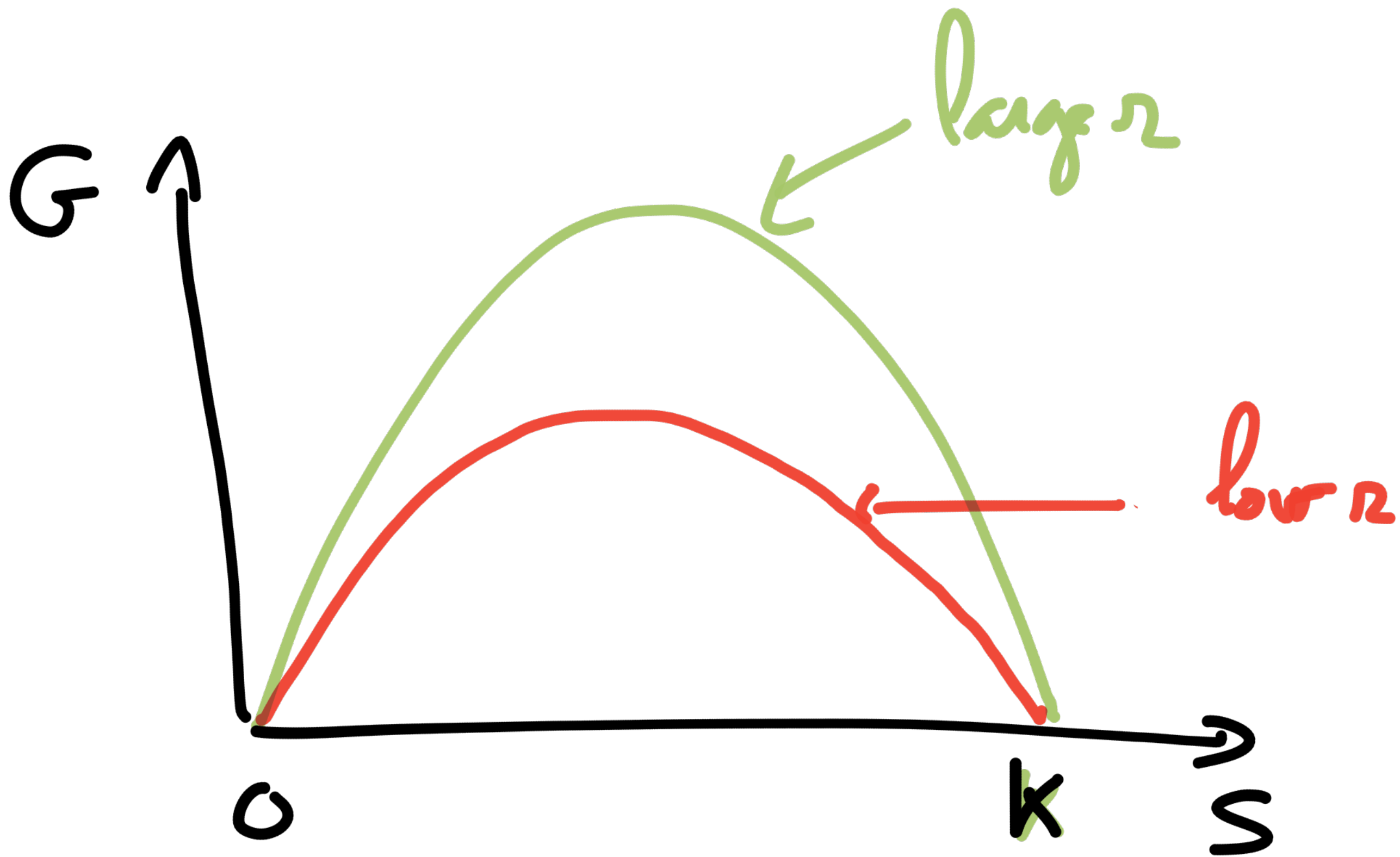


$$\dot{s}(t) = G(t)$$

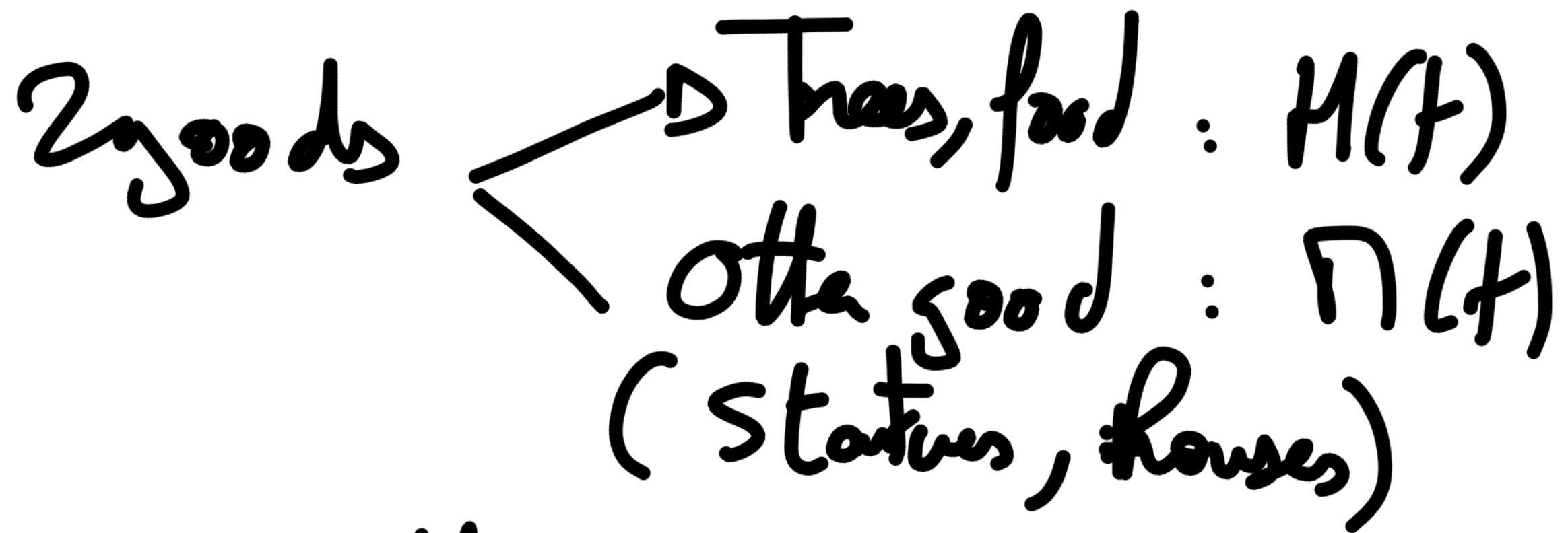


Assume $h=0$

("discrete time")



Economic model



• M is the numeraire

• P = price of trees, food

Technology

- only input = labor
- each living individual works 1 unit
- total labor input $L(t) = \text{pop. size}$

Harvest: $H^s(t) = \alpha S(t) L_h(t)$

Other good
(sharks)

$$H^s(t) = L_n(t)$$

(No capital)

- The resource $S(t)$ is free disposal
[nobody owns it]

Waglo:

$$w_H(t) = \frac{p(t) \dot{H}^3(t)}{L_H(t)} = p(t) \propto S(t)$$

$$w_n(t) = \frac{n^3(t)}{L_n(t)} = \underline{1}$$

Preferences

- individuals live only 1 period
- Fertility decisions are not derived from utility maximization.
- how people choose $\begin{cases} m(t) \\ h(t) \end{cases}$

Take one individual

$$u = m(t) h(t)^\beta; \quad 0 < \beta < 1$$

budget constraint

$$1 \times m(t) + p(t) h(t) = w(t)$$

by arbitrage $w_H(t) - w_N(t) = w(t)$

Solve $\max_{m, h} h(t)^\beta m(t)^{1-\beta}$

s.t. $m(t) + p(t)h(t) = w/A$

$$\mathcal{L} = h(t)^\beta m(t)^{1-\beta} + \lambda (w(t) - m(t) - p(t)h(t))$$

FOC (1) $\frac{\partial \mathcal{L}}{\partial m} = 0 \Leftrightarrow (1-\beta) \left(\frac{h(t)}{m(t)} \right)^\beta = 1$

(2) $\frac{\partial \mathcal{L}}{\partial h} = 0 \Leftrightarrow \beta \left(\frac{h(t)}{m(t)} \right)^{\beta-1} = 1 + p(t)$

$$\frac{(2)}{(1)} : \frac{\rho}{\rho} \frac{m(t)}{h(t)} = p(t)$$

$$\Rightarrow m(t) = \frac{\rho}{\rho} p(t) h(t)$$

$$\underline{BC} \quad p(t) h(t) + m(t) = w(t)$$

$$\Leftrightarrow p(t) h(t) + \frac{\rho}{\rho} p(t) h(t) = w(t)$$

$$P(t)h(t)\left(1 + \frac{1-\beta}{\beta}\right) = w(t)$$

$$h(t) = \beta \frac{w(t)}{P(t)} = h^d(t)$$

$$m(t) = (1-\beta) w(t) = m^d(t)$$

↳ Aggregate demands

$$H^d(t) = h^d(t) \times L(t), \quad M^d(t) = m^d(t) \times L(t)$$

$$M^d(t) = (1-p) w(t) L(t)$$

$$H^d(t) = p \frac{w(t) L(t)}{p(t)}$$

Price of the resource

$a_{LH}(S)$ - amount of labor needed
To cut one tree

- with $L(t)$, one can cut $\alpha S(t)L(t)$
- to cut 1 tree, one needs $\frac{L(t)}{\alpha S(t)L(t)}$
units of labor $\rightarrow a_{LH}(S) = \frac{1}{\alpha S(t)}$

In equilibrium, because S is free,
 $p(t)$ will be equal to the unit
cost of production:

$$p(t) = w(t) a_{LH}(t)$$

$$p(t) = \frac{w(t)}{a_{SC}(t)}$$

(Zero profit)

Equilibrium (economic)

L supply $\dot{L}^s(t) = L(t)$

L demand: $\dot{L}^d(t) = \dot{M}^s(t) + a_{L_h}(t) \times \dot{H}^s(t)$

Trees demand (for): $\dot{H}^d(t) = \beta \frac{w(t) L(t)}{p(t)}$

Trees supply: $\dot{H}^s(t) = \alpha S(t) L_{+1}(t)$

$w_H(t) = \alpha S(t) p(t), w_n(t) = 1, w_H^d(t) = w_n(t)$

Solving

$$M^S(t) = h^d(t) \Leftrightarrow \propto S(t) L_A(t) = \left(\frac{w(t) L_A(t)}{P_A(t)} \right)$$

$$P(t) = \frac{w(t)}{\propto S(t)}$$


$$H(t) = \propto \underbrace{S(t) L_A(t)}_{\text{given}}$$

Forest dynamics:

$$\dot{S}(t) = r S(t) \left(1 - \frac{S(t)}{K}\right) - \underbrace{\alpha \beta S(t) A t}_{H(t)}$$